

BENJAMIN HOUDEBERT

Principal Software Engineer | Software Architect | Technical Leader

Lille, France | +33 6 62 99 33 53 | bhoudebert@gmail.com

[linkedin.com/in/benjamin-houdebert](https://www.linkedin.com/in/benjamin-houdebert)

PROFESSIONAL SUMMARY

Software architect and technical leader with 20+ years of experience designing, building, and operating distributed backend systems, cloud-native platforms, APIs, and data-intensive products. Combines hands-on engineering in Rust, Java, Haskell, Kotlin, TypeScript, and Python with architecture ownership, team leadership, and product delivery. Experienced in multi-tenant SaaS, event-driven systems, identity and access management, payments, KYC, e-commerce, financial services, blockchain, and AI/RAG applications. Known for turning complex business and regulatory requirements into secure, observable, maintainable systems.

CORE EXPERTISE

Architecture: Distributed systems, domain-driven design (DDD), hexagonal architecture, microservices, event-driven architecture, multi-tenancy, API-first platforms, resilience patterns

Backend: Rust, Java, Haskell, Kotlin, TypeScript/Node.js, Python, Spring Boot, Axum, Actix Web, Servant, SQLx

APIs & Integration: REST, OpenAPI, gRPC, Protocol Buffers, GraphQL, WebSockets, OAuth 2.0, OpenID Connect, MCP

Data & Messaging: PostgreSQL, Redis, Kafka, Cassandra, Neo4j, Couchbase, HBase, Solr, data modeling, migrations, semantic/vector search

Cloud & Platform: Kubernetes, Docker, Helm, AWS, GCP, Azure, GitOps, Argo CD, GitHub Actions, Jenkins, Nix/NixOS

Engineering Leadership: Technical strategy, architecture decisions, mentoring, code quality, delivery ownership, stakeholder alignment, agile delivery

Reliability & Security: Observability, auditability, CI/CD, authentication, authorization, cryptography, secure platform design, GDPR-aware architecture

PROFESSIONAL EXPERIENCE

Confidential Client | Lead Software Engineer / Architect

May 2026 – July 2026

- Designed and delivered a reusable Retrieval-Augmented Generation (RAG) platform that ingests, vectorizes, indexes, and pre-warms internal documentation for context-aware answers.
- Built provider-independent LLM orchestration and contextual retrieval using semantic search and vector distance, enabling reuse across business domains.
- Implemented a React/TypeScript PWA and backend services in Rust/Axum and Node.js, backed by PostgreSQL and containerized with Docker Compose and Nix.

Agent Ride | Founder & Software Architect

October 2025 – Present

- Designed and built an agentic motorcycle trip-planning platform that converts a natural-language request into an itinerary grounded in verified routing, weather, traffic, daylight, fuel, and road data.
- Orchestrated schema-validated LLM tool use and parallel scouting agents while deterministic services enforce rider, route, safety, and time constraints.
- Developed motorcycle-specific routing and ride-time models using speed limits and road curvature; added GPX/navigation exports, duplicate detection, ride ratings, session tracing, and model cost/quality benchmarks.
- Exposed the platform capabilities through a Model Context Protocol (MCP) server for remote use from AI development tools.
- Technology:** TypeScript, Node.js, SQLite, Anthropic API, MCP, OpenStreetMap/Overpass, Valhalla, Open-Meteo, TomTom, Docker, automated tests.

Confidential Client | Lead Full-Stack / Rust Engineer

October 2025 – February 2026

- Built a consumer platform for mountain-bike geometry, fit analysis, product comparison, and purchase decision support.
- Developed SVG geometry visualization, personal bike collections, shared-library synchronization, versioning, diff-based updates, import/export workflows, rider-profile tuning, voting, and event scheduling.
- Delivered a responsive PWA with a React Native/TypeScript frontend and Rust/Actix Web backend using SQLx and PostgreSQL.

ADEO | Technical Leader

May 2025 – September 2025

- Led architecture and delivery of a multi-tenant backend platform within a large-scale delivery management system used across multiple business units.
- Provided day-to-day technical leadership for five developers and one QA engineer; established domain models, service boundaries, engineering standards, and delivery practices.
- Designed document ingestion, validation, metadata enrichment, audit-ready retention, asynchronous orchestration, notification, and background-processing workflows.
- Embedded compliance, traceability, resilience, and observability into the platform design while aligning technical decisions with product and stakeholder goals.
- Technology:** Java, Spring Boot, PostgreSQL, Liquibase, Kafka, OpenAPI, Resilience4j, Vault, Kubernetes, Docker, Nix, GitHub Actions, Karate, WireMock, TypeScript, Vue.

Unbox Universe | Head of Development / Principal Backend Engineer

March 2022 – May 2025

- Owned backend architecture and hands-on implementation for a modular platform while leading four teams across mobile, SaaS, platform, and blockchain products.
- Designed API-first services for payments, KYC, messaging, image moderation, digital receipts, and identity, using REST, gRPC, GraphQL, WebSockets, and event-driven processing.
- Built secure authentication and cryptographic workflows, including an OAuth 2.0-compatible authorization platform designed around passwordless identity.
- Developed scalable asynchronous services and operated them on Kubernetes through automated GitOps and CI/CD pipelines.
- Contributed to an open-source governance platform for non-sovereign funds and closed-loop networks, including Cardano smart contracts and Rust-to-Flutter integration.
- Technology:** Rust, Axum, Actix Web, Tokio, Haskell, Plutus/Plutarch, TypeScript, Node.js, PostgreSQL, Redis, Kafka, IPFS, Kubernetes, Argo CD, AWS, Nix.

Selected Consulting Engagements | Software Architect / Lead Engineer

2019 – 2023

Confidential Financial Platform — Lead Java Engineer (2021–2022)

Architected a unified application for managing investments distributed across providers. Built Java/Kotlin and Spring Boot services integrating PostgreSQL, Neo4j, Cassandra, Kafka, Matrix, and a Flutter client.

ADAFin — Haskell / Plutus Engineer (2021–2022)

Developed Cardano smart contracts for decentralized staking and token swaps, supported by Haskell services, PostgreSQL, Kafka, Kubernetes, and public-cloud infrastructure.

Decathlon — Haskell Software Engineer (2021)

Built services that ingest, normalize, and expose multinational product catalog, taxonomy, and attribute data through APIs and Kafka messaging. Used Haskell, Servant, Polysemy, PostgreSQL/JSONB, Nix, Kubernetes, and GCP.

North Devs — Software Architect / Cloud Engineer (2019–2023)

Architected Rust and Haskell platforms for moderated digital allowance and purchases, collaborative task management, and fitness planning. Delivered backend APIs, web/mobile clients, authentication, data services, and containerized infrastructure.

Cooptalis — Platform Technical Lead (2020)

Designed a central data platform that reconciled, versioned, and redistributed records across SaaS systems using configurable master/slave ownership and conflict-resolution rules. Also delivered a dynamic relocation workflow platform. Used Java, Spring Boot, Vavr, Kafka, PostgreSQL, Keycloak, Vue, Docker, Kubernetes, and GCP.

EARLIER EXPERIENCE

Malakoff Humanis | Technical Leader

October 2019 – December 2019

- Modernized CI/CD and deployment practices with Jenkins pipelines, Docker, automated testing, documentation-as-code, team notifications, and Consul; coached development and operations teams.

Decathlon France | Technical Leader — CRM & Identity

October 2017 – September 2019

- Led development of “Sign in with Decathlon,” an OAuth 2.0/OpenID Connect identity platform for customers and employees.
- Designed a distributed customer-data architecture with jurisdiction-aware GDPR controls, cross-data-center routing and synchronization, process-level data-usage traceability, and resilient APIs.
- Technology:** Java, Spring Boot/Cloud, Vavr, Kafka, Cassandra, Resilience4j, OpenTracing, Docker, Jenkins, Rancher.

Decathlon France | Technical Leader — Omnichannel Commerce

May 2015 – May 2017

- Helped deliver a multinational omnichannel commerce platform supporting consistent customer experiences across stores, web, and mobile.
- Technology:** Java, Scala, Play Framework, Vert.x, Couchbase, Oracle Commerce, Docker, Jenkins.

Groupe ADEO | Product Owner / Project Manager

December 2013 – April 2015

- Owned the backend product foundation for e-merchandising across multiple business units, including catalog ingestion, product families, ranking, boosting, variants, facets, and filters.
- Led a five-person team and coordinated development, UAT, production deployment, training, and rollout of the platform in Ukraine and Italy.
- Technology:** Java, REST, HBase, Solr/Cloudera Search, JMeter, Gatling, FitNesse.

Leroy Merlin | Team Manager / Technical Leader

December 2012 – October 2013

- Established and managed a four-person development service center responsible for 40 production applications, introducing Scrum/Kanban practices and service-level agreements.
- Led architecture for customer support, room-booking, store inventory, and mobile applications across Java, Scala, PHP, JavaScript, Android, and iOS stacks.

Cofidis | Development Coordinator / Lead Developer

January 2008 – November 2012

- Coordinated up to five concurrent projects, 20 developers, and initiatives of up to 150 person-days in consumer-credit and payment systems.
- Led enhancements for loan management, payment and regulatory data streams, and delivery monitoring using Java, Spring, SQL, Tomcat/WebLogic, and Jenkins.

Monabanq | Junior Project Manager / Software Developer

June 2007 – December 2008

- Managed projects of up to four people and 120 person-days; developed Monascope, an early expense-categorization feature for online banking, plus trading, savings, survey, document, and mortgage tools.

Leroy Merlin | Junior J2EE Architect

January 2007 – May 2007

- Built a reusable invoice/document-generation toolkit and authentication web services for stores using Java, Maven, LDAP, JBoss, Tomcat, and SQL.

SNCF | Software Developer

August 2005 – December 2006

- Developed railway geometry and maintenance applications, including a nightly ingestion pipeline processing 150 GB of infrastructure data.

EDUCATION

Master's Degree in Computer Science — Université d'Artois, Lens, France
2000 – 2005 | Software engineering, systems design, and artificial intelligence

LANGUAGES

French | English